

From: Brady, Donal (EGLE)
Sent: 10/17/2023 11:17:14 AM
To: "Braum, Thomas (EGLE)" <BraumT2@michigan.gov>; "Staron, Matthew (EGLE)" <STARONM@michigan.gov>
Cc: "Lowe, Claire (EGLE)" <LoweC9@michigan.gov>
Subject: RE: Cadillac WWTP NH4 Treatment Capacity

Hi Tom,

OK. Thanks!



MICHIGAN DEPARTMENT OF
ENVIRONMENT, GREAT LAKES, AND ENERGY

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From: Braum, Thomas (EGLE) <BraumT2@michigan.gov>
Sent: Tuesday, October 17, 2023 9:46 AM
To: Brady, Donal (EGLE) <BradyD6@michigan.gov>; Staron, Matthew (EGLE) <STARONM@michigan.gov>
Cc: Lowe, Claire (EGLE) <LoweC9@michigan.gov>
Subject: RE: Cadillac WWTP NH4 Treatment Capacity

Hi Donal,

There was new ammonia criterion developed in 2019 that we are beginning to incorporate into our permits. The water quality values for ammonia have been revised, resulting in revised WQBELs. When Phillip models for conventional pollutants, this new criterion has resulted in more stringent limits for some treatment plants. The attached memo goes into more detail about the background and our implementation plan.

Tom Braum
Environmental Quality Analyst
Municipal Permits Unit
Water Resources Division
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BraumT2@michigan.gov | 517-331-7377

From: Brady, Donal (EGLE) <BradyD6@michigan.gov>
Sent: Tuesday, October 17, 2023 9:30 AM
To: Braum, Thomas (EGLE) <BraumT2@michigan.gov>; Staron, Matthew (EGLE) <STARONM@michigan.gov>

Cc: Lowe, Claire (EGLE) <LoweC9@michigan.gov>
Subject: RE: Cadillac WWTP NH4 Treatment Capacity

Hi Tom,

Thanks, that explanation helps. The last question I have is why the ammonia limits changed for some plants, was this due to improved understanding of NH4 fate/transport/river dynamics vs. a water quality standard update?



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From: Braum, Thomas (EGLE) <BraumT2@michigan.gov>
Sent: Tuesday, October 17, 2023 8:31 AM
To: Staron, Matthew (EGLE) <STARONM@michigan.gov>; Brady, Donal (EGLE) <BradyD6@michigan.gov>
Cc: Lowe, Claire (EGLE) <LoweC9@michigan.gov>
Subject: RE: Cadillac WWTP NH4 Treatment Capacity

Hi Donal and Claire,

Your understanding is correct. Philip's analysis determined that Cadillac could currently meet the more stringent limits; however, because they are not discharging at their design flow, we initially included a certification condition in the permit that allowed Cadillac to make a determination whether they could meet the limits at their design capacity.

Thank you for looking into their aeration capacity. Let us know if you have additional questions.

Thanks,

Tom Braum
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From: Staron, Matthew (EGLE) <STARONM@michigan.gov>
Sent: Monday, October 16, 2023 7:23 AM
To: Brady, Donal (EGLE) <BradyD6@michigan.gov>; Braum, Thomas (EGLE) <BraumT2@michigan.gov>

Cc: Lowe, Claire (EGLE) <LoweC9@michigan.gov>
Subject: RE: Cadillac WWTP NH4 Treatment Capacity

Hello.

We provided the facility with an opportunity to certify that they could meet the new lower ammonia limits at design flow or indicate they need an SOC to comply if upgrades were needed. They declined to submit one or include it in the permit. Tom can provide additional details.

From: Brady, Donal (EGLE) <BradyD6@michigan.gov>
Sent: Sunday, October 15, 2023 2:15 PM
To: Braum, Thomas (EGLE) <BraumT2@michigan.gov>; Staron, Matthew (EGLE) <STARONM@michigan.gov>
Cc: Lowe, Claire (EGLE) <LoweC9@michigan.gov>
Subject: Cadillac WWTP NH4 Treatment Capacity

Hi Tom and Matt,

Claire and I are looking at the Cadillac WWTP's treatment capacity in completing some IPP related tasks. I wanted to double check with you regarding the concern that came up during the WWTP's permit renewal.

It looks like Phil D. evaluated MI WWTP capacities to meet new/lower ammonia limits based on past actual effluent quality and a potential to exceed analysis. I understand there was a concern some WWTPs would not be able to meet the lower limits based on the potential to exceed. I understand there was a concern other WWTPs would not be able to meet the lower limits, not because of the potential to exceed analysis, but because the actual plant flows were less than design flows. I understand the Cadillac WWTP was in the second group.

Can you please confirm my understanding about permit staff's concern with the Cadillac WWTP's NH4 treatment capacity is correct? I just want to make sure the concern wasn't also based on some further analysis. FYI, I looked at the Cadillac WWTP's aeration capacity and it seems sufficient for the design flow and NH4 loading.

Thanks!



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